READING NOTE #002

Multi-Embed: Unified Morphology-Molecule Representation Learning

01. Paper Information

Paper	Systematically decoding pathological morphologies and molecular profiles with unified multimodal embedding
Venue	Nature Methods, 2026
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Table 1: At a glance summary of the Multi-Embed paper.

Aspect	Reading note
Question	Can pathology morphology and multilayer molecular profiles be aligned into one interpretable embedding space for cross-modality inference, integration, prognosis, and spatial dynamics?
Data	Image-omics matched data at multiple scales: TCGA/CPTAC slide-level cohorts, multiple spatial transcriptomics platforms, spatial proteomics, and external clinical cohorts.
Model	UNI-derived morphology features and molecular profiles are projected into a shared Multi-Embed space with reconstruction, self-supervised triplet alignment, and representation regularizers.
Output	Unified morphology-molecule embeddings, image-to-molecule prediction, spatial clustering, prognosis prediction, TLS identification, and malignant pseudo-trajectory analysis.
Main claim	A unified embedding can jointly support cross-modality inference and integration across bulk, spot-level, and single-cell-level pathology-omics data.

Reading map

1 One-sentence summary

Core summary

This paper proposes **Multi-Embed**, a self-supervised multimodal learning framework that maps multilevel pathological morphologies and multilayer molecular profiles into a unified embedding space, enabling morphology-to-molecule inference, multimodal integration, prognostic modeling, TLS identification, and malignant spatiotemporal trajectory analysis.

2 Why this paper matters

Multi-Embed tries to move beyond a single clinical endpoint. Instead of only asking whether H&E plus omics improves prognosis, it asks a broader representation-learning question: can one shared morphology-molecule embedding support **multiple downstream tasks across multiple data scales**?

This makes the paper conceptually different from PORPOISE. PORPOISE focuses on supervised multimodal prognosis with bulk omics, while Multi-Embed first learns a **cross-modality representation** and then reuses it for inference and integration. It also connects naturally to later H&E-ST work such as STORM, although Multi-Embed is better viewed as a unified image-omics embedding framework than a single giant spatial foundation model.

Key takeaway

The main value of this paper is its framing: pathology-omics learning is not only a patient-level fusion problem, but also a cross-scale alignment problem. A useful representation should connect slide-level bulk omics, spot-level spatial transcriptomics, single-cell-level spatial data, and multilayer molecular profiles within one embedding framework.

3 Problem formulation

The paper studies **cross-modality inference and integration** between pathology morphology and molecular profiles. The available inputs can be organized at several levels:

- **Slide-level / patient-level data**: H&E whole-slide images paired with bulk molecular profiles from TCGA and CPTAC.
- **Spot-level spatial data**: spatial transcriptomics spots paired with local H&E tissue regions.
- **Single-cell-level spatial data**: high-resolution spatial platforms where image patches can be paired with cellular or small-bin molecular measurements.
- **Multilayer molecular profiles**: genomics, epigenomics, transcriptomics, and proteomics.

The model learns a shared embedding in which paired morphology and molecular features should be close, while each modality should still retain enough information for reconstruction and downstream inference.

Core modeling assumption

Assumption 3.1

The paper assumes that pathological morphology and molecular profiles contain both **shared disease-state information** and **modality-specific complementary information**. If the model can align paired morphology and molecular samples without collapsing either modality, the learned space should support both cross-modality prediction and biological discovery.

4 Method breakdown

4.1 Overall architecture

The Multi-Embed workflow has four main parts:

1. **Morphology feature extraction**: pathology tiles are encoded using the UNI pathology foundation model.
2. **Molecular feature preparation**: normalized molecular profiles are used as input features, including expression, methylation, mutation, and protein abundance.
3. **Projection into Multi-Embed space**: modality-specific MLP encoders map morphology and molecular features into latent representations, followed by a shared projection layer.
4. **Cross-modality learning**: reconstruction loss preserves intra-modality information, while triplet-based self-supervised loss aligns paired morphology and molecular features.

Method note

The method is not simply feature concatenation. It first creates a cross-modality embedding space, then evaluates whether that space can support many tasks: molecular prediction from H&E, spatial clustering, prognosis, TLS detection, and trajectory analysis. This is the key difference from a task-specific multimodal survival model.

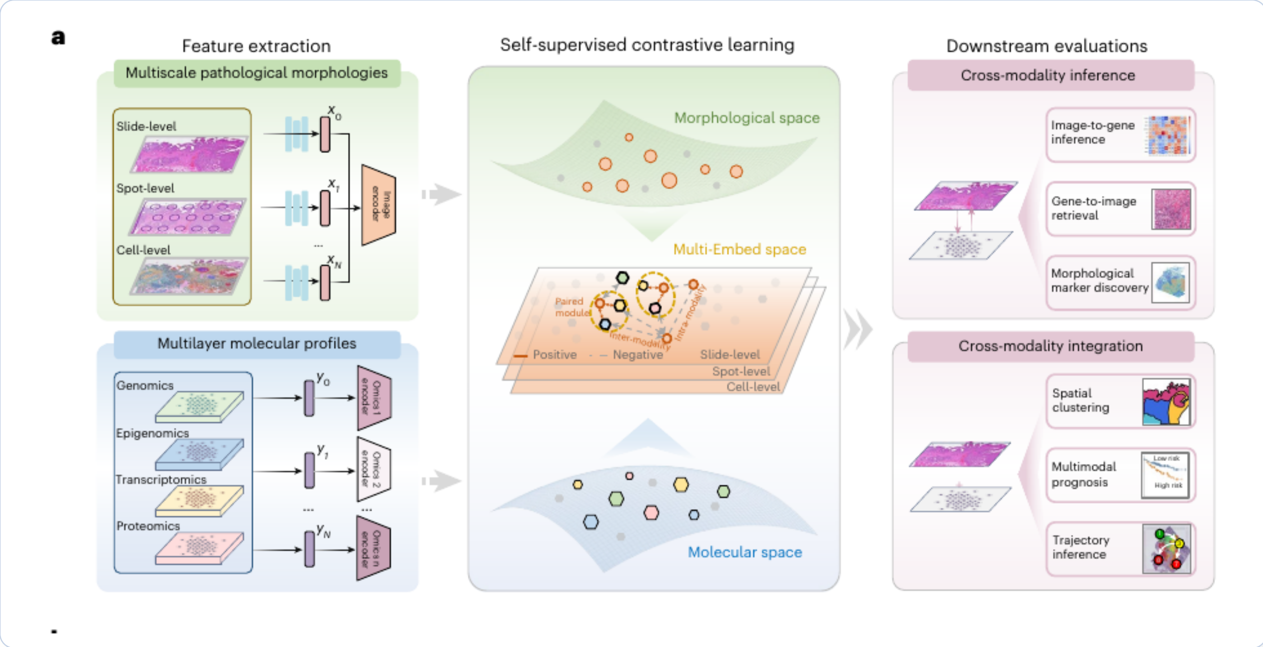

Figure 1: Figure 1a of the paper. Multi-Embed extracts multiscale pathology morphology and multilayer molecular features, aligns them through self-supervised contrastive learning, and uses the learned embedding for cross-modality inference and integration.

4.2 Loss design

The training objective combines four terms:

Table 2: Main training signals used by Multi-Embed.

Loss term	Role	Interpretation
Reconstruction loss	Preserves modality information	Morphology and molecular embeddings should reconstruct their original inputs rather than becoming overly compressed labels.
Self-supervised triplet loss	Aligns paired modalities	Paired image-omics samples are pulled closer than unpaired samples in the shared space.
Divergence regularizer	Avoids representation collapse	Penalizes overly similar element slots so that the learned representations remain diverse.
MMD regularizer	Stabilizes early alignment	Encourages distributions of morphology and molecular embeddings not to diverge too strongly during training.

For slide-level data, a WSI is a set of many morphology tile embeddings paired with one bulk molecular profile. Multi-Embed therefore uses a set-based weakly supervised triplet loss with smooth-Chamfer similarity. For spot-level and single-cell-level data, the pairing is more direct: each local image tile corresponds to one spatial molecular measurement.

Why the set-based loss matters

Method 4.1

Bulk RNA-seq is not spatially localized to each tile. Treating a WSI as a set avoids pretending that every tile has a precise molecular label. The weakly supervised set loss is therefore a practical bridge between slide-level bulk omics and tile-level morphology.

5 Main experimental results

How the evidence is organized in this note

Editorial note 5.1

To make the experimental section easier to read, I split the large composite figures in the paper into smaller visual units. Each subsection below contains a corresponding cropped result figure and a short interpretation, so that no experimental claim is left as text alone.

5.1 Bulk RNA inference across TCGA cancer types

The first benchmark asks whether H&E morphology can predict gene expression profiles on TCGA bulk cohorts. Multi-Embed reaches an overall average PCC of **0.37** for highly variable genes, compared with 0.29 for HE2RNA and 0.32 for DeepPT. The improvement is also visible cancer-by-cancer across the 12 TCGA tumor types shown in panel c.

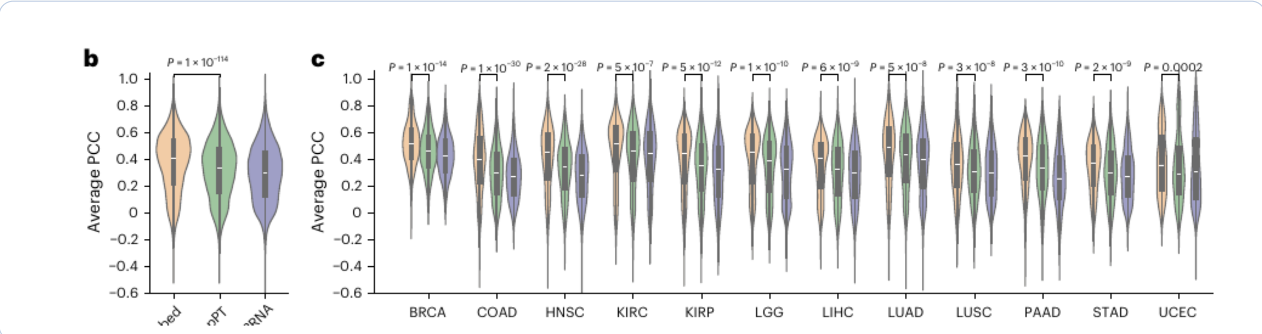

Figure 2: Figure 1b–c of the paper. Bulk morphology-to-expression inference on TCGA. Multi-Embed achieves the best overall and cancer-specific mean PCC when predicting highly variable gene expression from pathology images.

What this result means

Figure note 5.2

This is the clearest first piece of evidence that the embedding is biologically informative rather than merely visually smooth. If morphology-aligned embeddings can recover bulk RNA variation better than earlier models, they likely capture disease signals beyond superficial texture.

5.2 Spatial gene-expression inference across Visium, Xenium, and Visium HD

The same idea is then tested on spatial transcriptomics data. The paper reports higher mean PCC on 10x Visium, 10x Xenium, and 10x Visium HD datasets than the comparison methods iStar and

OmiCLIP. This is important because it shows that the framework is not restricted to bulk cohorts: it also transfers to spot-level and near-cell-level spatial settings.

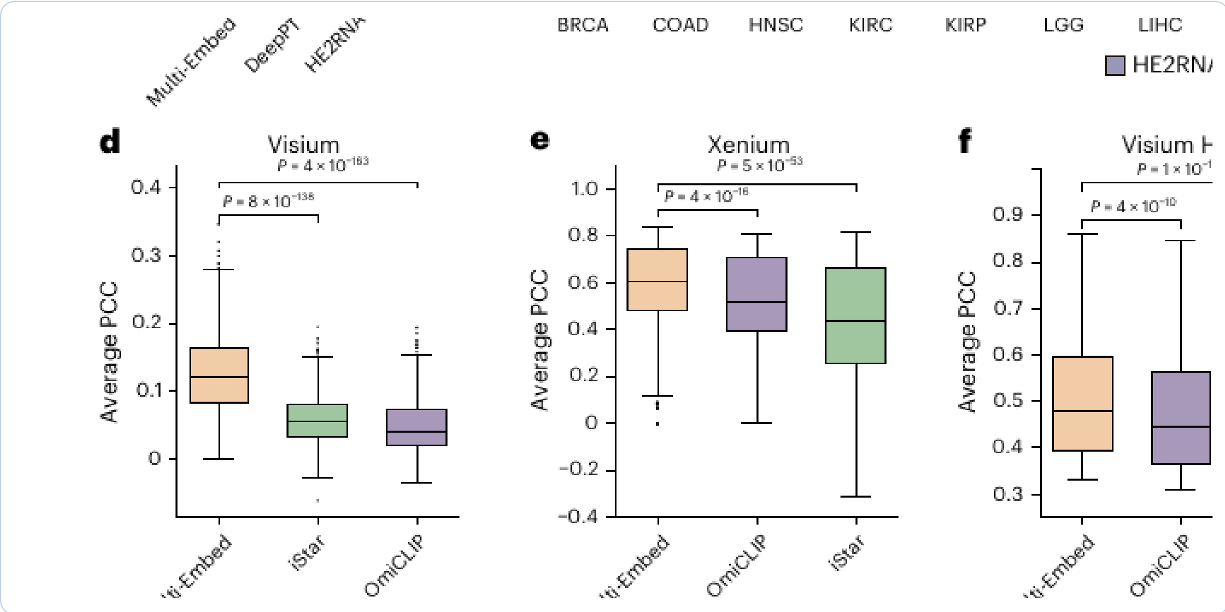

Figure 3: Figure 1d–f of the paper. Spatial morphology-to-expression inference across three platforms. Multi-Embed outperforms the comparison methods on Visium, Xenium, and Visium HD benchmarks.

My reading

Figure note 5.3

This subsection is where the **cross-scale generality** claim becomes credible. The model is not only good at bulk expression inference; it also retains enough locality to work on genuinely spatial assays.

5.3 Inference of other molecular layers: methylation, protein, mutation, and spatial proteomics

A particularly strong part of the paper is that it does not stop at RNA. Multi-Embed is also tested on image-to-methylation inference, image-to-protein inference, mutation-status prediction, and spatial proteomics prediction. Across these tasks it consistently beats the corresponding task-specific baselines.

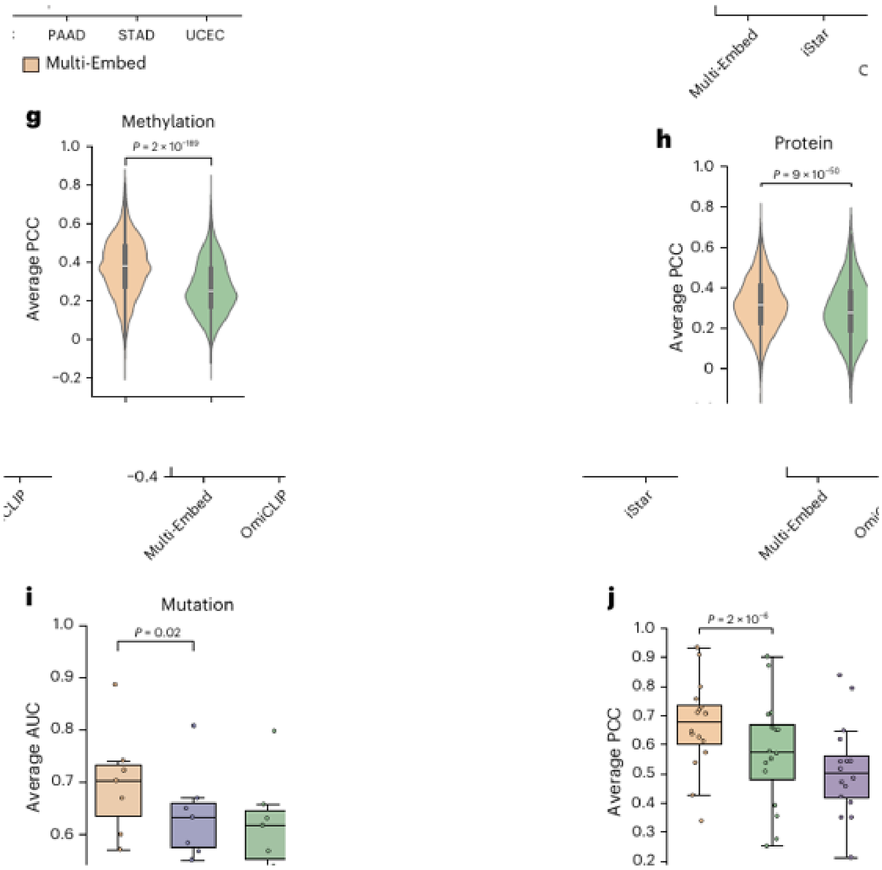

Figure 4: Figure 1g–j of the paper. Multi-Embed supports morphology-to-methylation, morphology-to-protein, morphology-to-mutation, and spatial proteomics inference. The figure strengthens the claim that the learned embedding generalizes across molecular layers rather than only RNA expression.

Why this figure matters

Figure note 5.4

This is one of the paper's most distinctive experimental points. The representation is intended to align *multilayer* molecular states with morphology. Showing evidence across methylation, protein abundance, and mutation status makes the "unified embedding" story much more convincing.

5.4 Cross-modality integration for spatial clustering

Besides cross-modality prediction, the authors also evaluate **cross-modality integration**. On the TNBC spatial clustering benchmark, Multi-Embed achieves the highest ARI and outperforms multi-modal baselines such as OmiCLIP, MISO, MUSE, and SpaGCN.

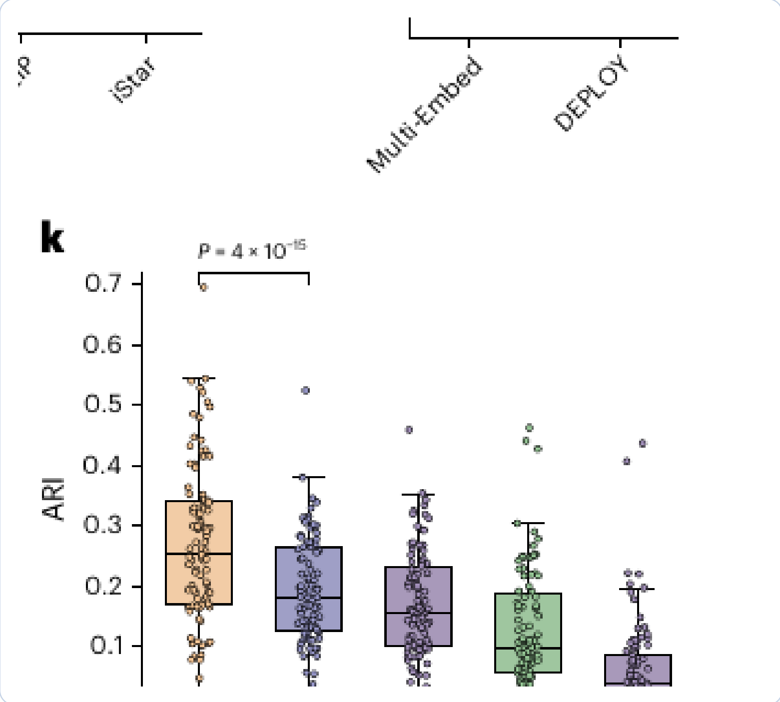

Figure 5: Figure 1k of the paper. Spatial clustering benchmark on the TNBC dataset. Multi-Embed achieves the highest ARI among the compared methods.

> **Interpretation**
>
> Figure note 5.5
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> This panel is important because it separates **prediction** from **integration**. A model might reconstruct one modality from another yet still fail to produce a useful joint tissue representation. Here, Multi-Embed performs well on both fronts.

5.5 Multimodal prognosis across 12 TCGA cancer types

The embedding is next used for multimodal prognosis prediction. Compared with PORPOISE-based multimodal prognosis models, Multi-Embed achieves significantly higher C-index values across the 12 TCGA cancer types in Figure 2a. The improvement is broad rather than limited to one or two datasets.

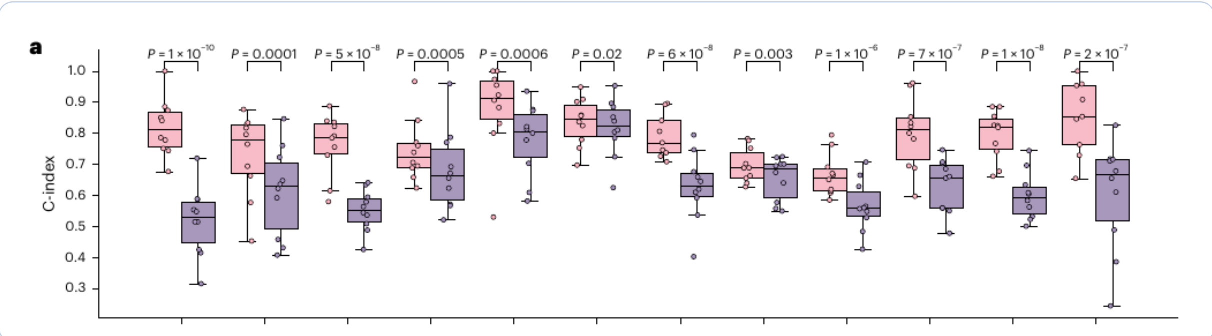

Figure 6: Figure 2a of the paper. Prognosis prediction across 12 TCGA cancer types. The Multi-Embed-based prognosis model consistently achieves higher C-index values than the PORPOISE-based comparison model.

Connection to PORPOISE

Figure note 5.6

This result suggests that learning a representation first and then using it for prognosis can be stronger than directly building a task-specific fusion model. It is also the cleanest bridge between Multi-Embed and the earlier PORPOISE paper.

5.6 Interpretable high-risk region and morphology-gene evidence

The prognosis section also includes an interpretability example. The paper shows one representative high-risk region that is emphasized by Multi-Embed but largely missed by the PORPOISE-based model, together with the morphology-inferred expression profile of the selected region.

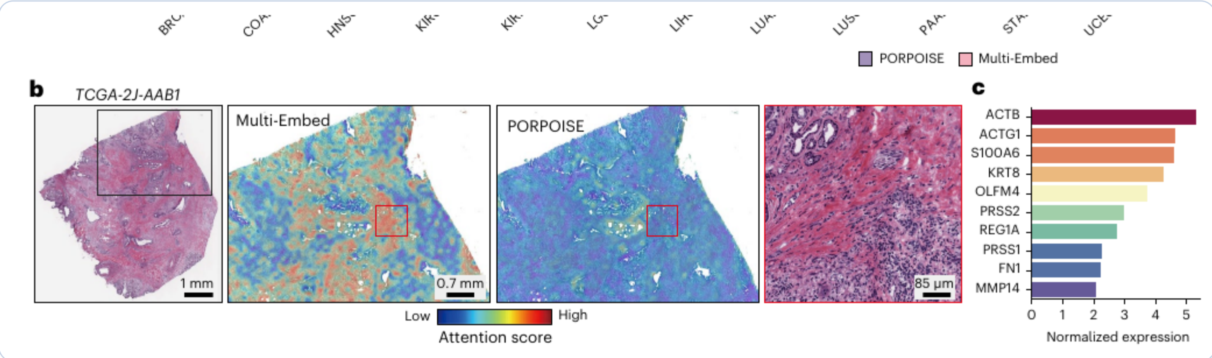

Figure 7: Figure 2b–c of the paper. A representative prognosis-risk example. Multi-Embed highlights a high-risk region and connects it to a corresponding molecular-expression signature, enabling joint image-omics interpretation.

Why I like this experiment

Figure note 5.7

This is where the model feels most useful biologically. The paper is not only claiming a better score; it is also attempting to show **which region** drives risk and **which genes** are associated with that region in the learned embedding space.

5.7 External prognosis validation

The prognosis claim is further tested on six external cohorts: TNBC, SurGen, CPTAC-HNSC, CPTAC-LUAD, CPTAC-PAAD, and CPTAC-UCEC. In all six datasets, the Multi-Embed-based prognosis model outperforms the PORPOISE-based comparator.

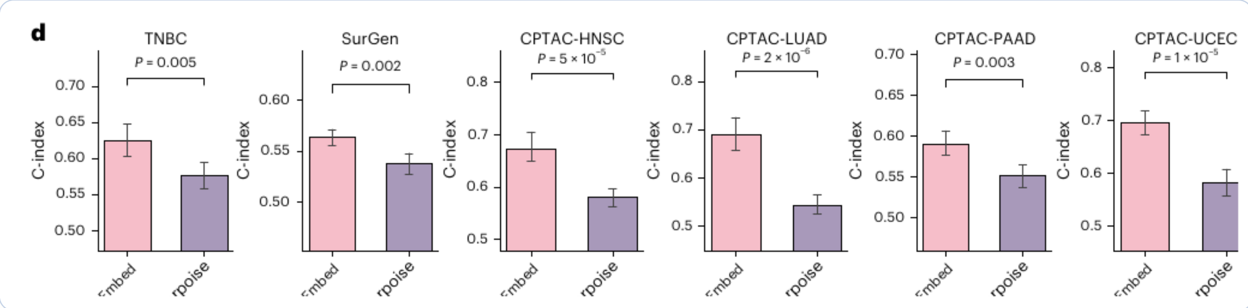

Figure 8: Figure 2d of the paper. External prognosis validation across six independent cohorts. Multi-Embed remains better than the PORPOISE-based baseline on each displayed dataset.

> **What this adds**
>
> Figure note 5.8
>
> External validation is critical here. Since many pathology-omics models improve on in-sample benchmarks but weaken under distribution shift, the consistency across all six cohorts makes the prognosis result substantially more credible.

5.8 TLS identification in TNBC

The paper also studies tertiary lymphoid structure identification in TNBC spatial data. Multi-Embed achieves higher F1 than OmiCLIP and MISO, and the qualitative map shows that it can identify additional TLS regions that were later confirmed by pathologists.

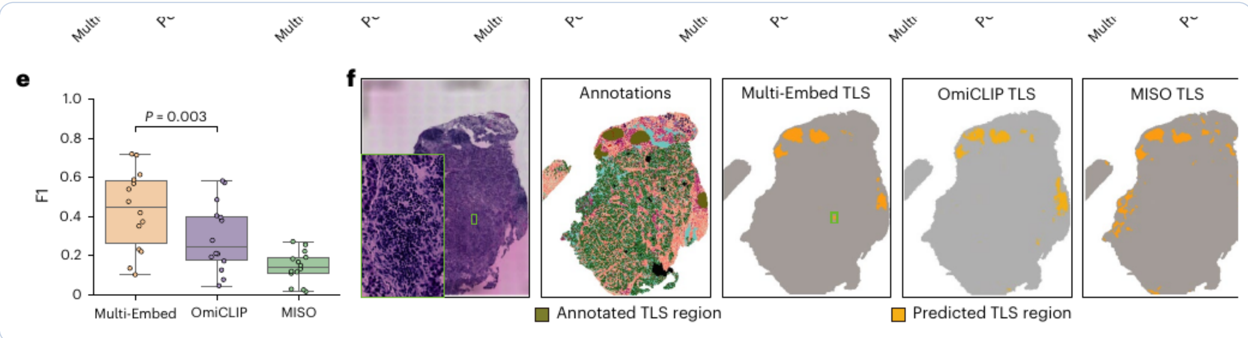

Figure 9: Figure 2e–f of the paper. TLS identification in TNBC. Multi-Embed achieves the best F1 score and also produces a qualitatively stronger TLS map than the comparison methods.

> **My interpretation**
>
> Figure note 5.9
>
> This is an excellent example of a **fine-grained tissue-architecture task**. It shows that the embedding is not only useful for global prediction, but also for discovering biologically meaningful spatial structures inside a tissue section.

5.9 Gastric lesion clustering and malignant pseudo-trajectory

Finally, the Multi-Embed space is used to organize stomach-cancer spatial transcriptomics data. The figure combines pathology-lesion annotation, spatial clustering, ARI comparison, pseudo-time ordering, and correlations between pseudo-time and both CNV score and an early-gastric-cancer signature.

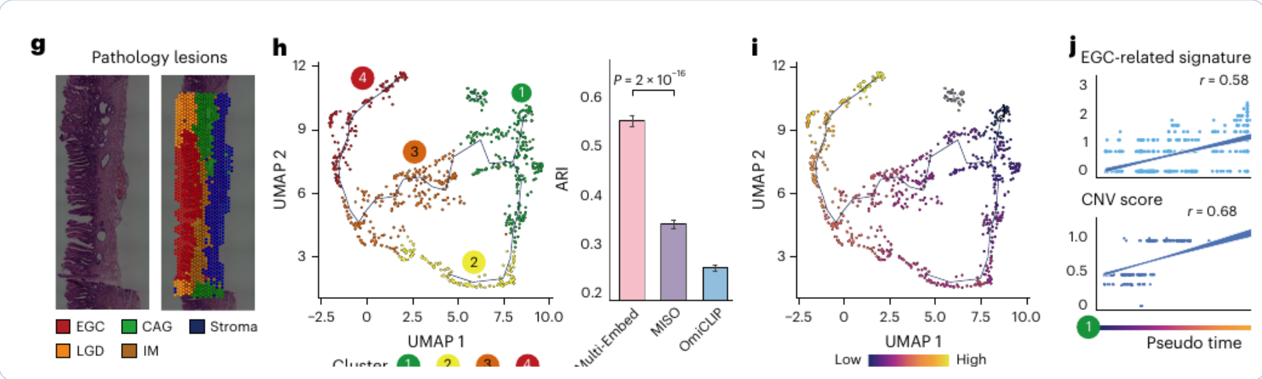

Figure 10: Figure 2g–j of the paper. Gastric-cancer spatial analysis. Multi-Embed supports lesion-aware clustering, yields a higher ARI than the comparison methods, and organizes spatial spots along a malignant pseudo-trajectory correlated with CNV score and an EGC-related signature.

> **Caution and value**
>
> Figure note 5.10
>
> This is probably the most ambitious application in the paper. I find it exciting because it links morphology, spatial omics, and disease progression in one view. At the same time, pseudo-time here should be interpreted carefully as an inferred *progression axis*, not direct longitudinal time.

5.10 What the experimental section shows overall

Taken together, the experiments support a coherent story:

- the embedding supports **cross-modality inference** (bulk RNA, ST RNA, methylation, protein, mutation, spatial proteomics),
- it supports **cross-modality integration** (spatial clustering),
- it can be transferred to **clinical tasks** (multimodal prognosis with internal and external validation), and
- it can drive **biological discovery tasks** (TLS identification and malignant pseudo-trajectory analysis).

> **Key takeaway**
>
> The strength of the experimental section is not one isolated metric. It is the fact that the same representation-learning framework is evaluated across many different task families, and each family has a corresponding visual result. This is exactly the kind of evidence a “unified embedding” paper needs.

6 What I learned

6.1 1. A good pathology-omics representation should support more than one task

The paper is valuable because it evaluates the learned representation across diverse tasks. This is a useful design lesson: if a representation is genuinely meaningful, it should transfer across prediction, integration, clinical modeling, and biological discovery.

6.2 2. Bulk and spatial data can be placed in one conceptual framework

Multi-Embed treats slide-level bulk omics and spot-level spatial omics as different pairing regimes. This is a practical idea for pathology-omics research because most clinical cohorts have bulk omics, while spatial omics is more local but usually smaller in scale.

6.3 3. Alignment is useful, but biological validation remains essential

Higher PCC, ARI, C-index, or F1 does not by itself prove biological mechanism. The most convincing parts are those where the embedding leads to interpretable regions, TLS discovery, CNV-associated trajectories, or marker-gene patterns that can be externally validated.

7 Relationship to PORPOISE and STORM

Table 3: How Multi-Embed relates to nearby pathology-omics models.

Paper	Main input	Core task	Main idea
PORPOISE	H&E WSI + bulk omics	Survival prognosis	Task-specific multimodal survival fusion with image attention and molecular attribution.
Multi-Embed	H&E + multilayer omics / spatial omics	Cross-modality inference and integration	Unified morphology-molecule embedding across bulk, spot-level, and single-cell-level data.
STORM	H&E + spatial transcriptomics	Spatial domain discovery, virtual ST, clinical prediction	Spatially aware H&E-ST foundation model with neighborhood context and virtual ST generation.

Multi-Embed sits between PORPOISE and STORM. Compared with PORPOISE, it is more representation-oriented and covers more molecular layers. Compared with STORM, it is less of a large pan-tissue spatial foundation model, but more general in the types of molecular profiles it tries to integrate.

Implications for my work

For my own pathology-omics work, this paper suggests a useful route: learn a molecularly grounded H&E representation first, then evaluate it on clinically meaningful tasks such as prognosis, response prediction, tissue structure discovery, and trajectory-like disease progression. The key is to show that alignment improves downstream biology, not just embedding metrics.

8 Critical reading

Critical reading

Strengths. The paper has a broad task suite, strong clinical motivation, multiple molecular layers, multi-scale data, and interpretable downstream applications. It also explicitly compares against both task-specific and multimodal baselines.
Limitations. Multi-Embed still depends on image-omics matched data, and the scale may be insufficient for a true plug-and-play foundation model. The trajectory analysis uses Monocle3 after representation learning, so the temporal inference is not fully integrated into the model. Some results may also depend on preprocessing, pairing assumptions, and the quality of molecular labels.
Evaluation concern. Because the paper spans many tasks, each individual benchmark receives limited space. For future work, it would be useful to have more detailed ablations of reconstruction, triplet loss, MMD, divergence regularization, and the set-based weak supervision strategy.

9 Reusable ideas for my own research

1. **Cross-scale pairing design.** Use different losses for bulk slide-level pairs and spatial spot-level pairs instead of forcing one pairing assumption.
2. **Representation-first evaluation.** Test whether a learned embedding improves multiple downstream tasks, not only a single prediction score.
3. **Embedding-to-biology loop.** Link high-attention or high-similarity regions back to genes, pathways, TLS regions, CNV patterns, and spatial domains.
4. **Comparison with task-specific fusion.** Compare a representation-learning approach against direct fusion baselines such as PORPOISE to show why pre-alignment helps.

10 Open questions

Questions for further thought

- How stable is the Multi-Embed space across different pathology foundation encoders, such as UNI, Virchow, GigaPath, or newer pathology models?
- Can the lagged or dynamic relationships between molecular layers be modeled directly rather than inferred through post hoc trajectory tools?
- How much performance comes from the shared embedding itself versus the strong UNI morphology encoder?
- Can this framework be extended to partially paired or mostly unpaired pathology-omics cohorts?
- Are the inferred morphology-molecule associations causal, prognostic, or simply correlated with cancer type and tissue composition?

11 Final takeaway

Multi-Embed is a strong example of how pathology-omics research is shifting from task-specific fusion toward reusable multimodal representation learning. Its key contribution is not one single benchmark, but the attempt to place morphology, bulk omics, spatial omics, prognosis, TLS discovery, and malignant trajectory analysis under a unified embedding framework. For future work, the most important next step is to make such representations more robust, more interpretable, and more directly connected to dynamic biological mechanisms.